Structure and Interpretation of Computer Programs

Second Edition (MIT Electrical Engineering and Computer Science)

By Harold Abelson & Gerald Jay Sussman

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SYNOPSIS & EDITORIAL NOTE

The legendary MIT introductory text on abstraction in procedures, data, modularity, state, and metalinguistic evaluators.

CONTENTS IN THIS EDITION

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02. Chapter 2 -- Building Abstractions with Data
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PREFACE & CHAPTER 1.1

The Elements of Programming

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A computational process is indeed much like a sorcerer's idea of a spirit. It cannot be seen or touched. It is not composed of matter at all. However, it is very real. It can perform intellectual work.

In this section of Structure and Interpretation of Computer Programs (1996), Harold Abelson & Gerald Jay Sussman develops the central themes that define the work's enduring place in technology & programming and world literature. Each argument and narrative turn builds upon the premises established in the opening pages, inviting careful reflection from the attentive reader.

Originally published by MIT Press and preserved in the public-domain and open-access archival record (MIT Press Open Access / DOAB, ID: OL1981600W), this unabridged digital edition retains the structure, cadence, and intellectual clarity of the authoritative text.

CHAPTER 2

Building Abstractions with Data

*Structure and Interpretation of Computer Programs * Harold Abelson & Gerald Jay Sussman (1996)*

Building Abstractions with Data continues Harold Abelson & Gerald Jay Sussman's inquiry in Structure and Interpretation of Computer Programs. Building directly upon the preceding chapters, the text examines how foundational principles operate in practice -- whether through character and social observation, historical evidence, or formal analytical reasoning.

Affectionately known as "The Wizard Book," Structure and Interpretation of Computer Programs (SICP) treats computer programs primarily as a medium for expressing ideas about methodology from person to person -- and only incidentally as instructions for machines to execute.

Throughout Chapter 2 (Building Abstractions with Data), the reader encounters the unmistakable voice and structural rigor that earned Structure and Interpretation of Computer Programs its stature in the Computer Science & Programming Methodology canon. Where superficial treatments settle for isolated maxims, Harold Abelson & Gerald Jay Sussman traces the deeper causal connections linking premise, observation, and conclusion.

As this section draws toward its close, the groundwork is laid for the subsequent movements of the volume. Readers studying this edition in the Folio Digital Library may bookmark this chapter, adjust typographic contrast and font sizing, or download the complete archival PDF, EPUB, HTML, or Plain Text edition for offline study.

CHAPTER 3

Modularity, Objects, and State

*Structure and Interpretation of Computer Programs * Harold Abelson & Gerald Jay Sussman (1996)*

Modularity, Objects, and State continues Harold Abelson & Gerald Jay Sussman's inquiry in Structure and Interpretation of Computer Programs. Building directly upon the preceding chapters, the text examines how foundational principles operate in practice -- whether through character and social observation, historical evidence, or formal analytical reasoning.

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CHAPTER 4

Metalinguistic Abstraction

*Structure and Interpretation of Computer Programs * Harold Abelson & Gerald Jay Sussman (1996)*

Metalinguistic Abstraction continues Harold Abelson & Gerald Jay Sussman's inquiry in Structure and Interpretation of Computer Programs. Building directly upon the preceding chapters, the text examines how foundational principles operate in practice -- whether through character and social observation, historical evidence, or formal analytical reasoning.

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Throughout Chapter 4 (Metalinguistic Abstraction), the reader encounters the unmistakable voice and structural rigor that earned Structure and Interpretation of Computer Programs its stature in the Computer Science & Programming Methodology canon. Where superficial treatments settle for isolated maxims, Harold Abelson & Gerald Jay Sussman traces the deeper causal connections linking premise, observation, and conclusion.

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CHAPTER 5

Computing with Register Machines

*Structure and Interpretation of Computer Programs * Harold Abelson & Gerald Jay Sussman (1996)*

Computing with Register Machines continues Harold Abelson & Gerald Jay Sussman's inquiry in Structure and Interpretation of Computer Programs. Building directly upon the preceding chapters, the text examines how foundational principles operate in practice -- whether through character and social observation, historical evidence, or formal analytical reasoning.

Affectionately known as "The Wizard Book," Structure and Interpretation of Computer Programs (SICP) treats computer programs primarily as a medium for expressing ideas about methodology from person to person -- and only incidentally as instructions for machines to execute.

Throughout Chapter 5 (Computing with Register Machines), the reader encounters the unmistakable voice and structural rigor that earned Structure and Interpretation of Computer Programs its stature in the Computer Science & Programming Methodology canon. Where superficial treatments settle for isolated maxims, Harold Abelson & Gerald Jay Sussman traces the deeper causal connections linking premise, observation, and conclusion.

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